Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
.Net Core and Progressive Web Application Practical

Practical – II

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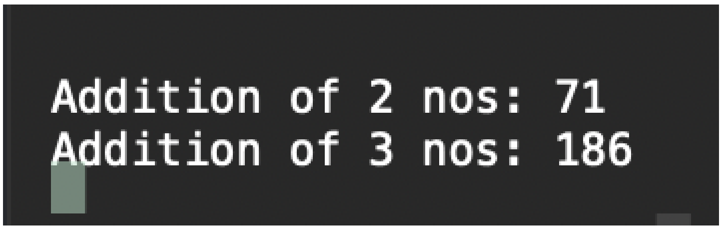
AIM: Write the program with different features of C#.

a. **Function Overloading.**

CODE:

```
using System;

public class MethodOverloading
{
   void Function(int n1, int n2)
   {
      Console.WriteLine("Addition of 2 nos: " + (n1 + n2));
   }

   void Function(int n1, int n2, int n3)
   {
      Console.WriteLine("Addition of 3 nos: " + (n1 + n2 + n3));
   }

   public static void Main(string[] args)
   {
      MethodOverloading obj = new MethodOverloading();

      obj.Function(43, 28);
      obj.Function(21, 82, 83);
   }
}
```

OUTPUT:



b. Inheritance (all types).

CODE: (" : " is used to inherit class)

1. Single Inheritance

```
using System;

public class A
{
    public void function1()
    {
        Console.WriteLine("This is method of class A.");
    }
}

public class B : A
{
    public void function2()
    {
        Console.WriteLine("This is method of class B.");
    }
}

class SingleInheritance
{
    public static void Main(string[] args)
    {
        B obj = new B();

        Console.WriteLine("=======Single Inheritance=======");
        obj.function1();
        obj.function2();
        Console.WriteLine("Himani T016");
    }
}
```

2. Multilevel Inheritance

```
using System;

public class A
{
    public void function1()
    {
        Console.WriteLine("This is method of class A.");
    }
}

public class B : A
{
    public void function2()
    {
        Console.WriteLine("This is method of class B.");
    }
}

public class C : B
{
    public void function3()
    {
        Console.WriteLine("This is method of class C.");
    }
}

class MultilevelInheritance
```

```
{
   public static void Main(string[] args)
   {
      Console.WriteLine("==== Multilevel Inheritance ====");

      C obj1 = new C();

      obj1.function1();
      obj1.function2();
      obj1.function3();
      Console.WriteLine("Himani T016");
   }
}
```

3. **Hierarchical Inheritance**

```
using System;

public class A
{
   public void function1()
   {
      Console.WriteLine("This is method of class A.");
   }
}

public class B : A
{
   public void function2()
   {
      Console.WriteLine("This is method of class B.");
   }
}

public class C : B
{
   public void function3()
   {
      Console.WriteLine("This is method of class C.");
   }
}

class MultilevelInheritance
{
   public static void Main(string[] args)
   {
      Console.WriteLine("==== Hierarchical Inheritance ====");

      C obj1 = new C();

      obj1.function1();
      obj1.function2();
      obj1.function3();
      Console.WriteLine("Himani T016");
   }
}
```

D. Multiple Inheritance (using Interface)

```
using System;

interface IA
{
    public void functionA();
}

interface IB
{
    public void functionB();
}
class C : IA, IB
{
    public void functionA()
    {
        Console.WriteLine("This is method of Interface A.");
    }

    public void functionB()
    {
        Console.WriteLine("This is method of Interface B.");
    }
}

class MultipleInheritance
{
    public static void Main(string[] args)
    {
        Console.WriteLine("===== Multiple Inheritance =====");

        C obj = new C();
        obj.functionA();
        obj.functionB();
        Console.WriteLine("Himani T016");
    }
}
```

```
=======Single Inheritance=======
This is method of class A.
This is method of class B.
Himani T016
```

```
==== Multilevel Inheritance ====
This is method of class A.
This is method of class B.
This is method of class C.
Himani T016
```

```
==== Hierarchical Inheritance ====
This is method of class A.
This is method of class B.
This is method of class C.
Himani T016
```

```
===== Multiple Inheritance =====
This is method of Interface A.
This is method of Interface B.
Himani T016
```

c. Constructor overloading

CODE:

```
using System;

public class MethodOverloading
{
   MethodOverloading(int n1, int n2)
   {
      Console.WriteLine("Addition of 2 nos: " + (n1 + n2));
   }

   MethodOverloading(int n1, int n2, int n3)
   {
      Console.WriteLine("Addition of 3 nos: " + (n1 + n2 + n3));
   }

   public static void Main(string[] args)
   {
      MethodOverloading obj = new MethodOverloading(43, 28);
      MethodOverloading obj1 = new MethodOverloading(43, 28, 37);
      Console.WriteLine("Himani T016");
   }
}
```

OUTPUT:

```
Addition of 2 nos: 71
Addition of 3 nos: 108
Himani T016
```

E. Using Delegates and events

CODE:

```
using System;

public class MethodOverloading
{
    MethodOverloading(int n1, int n2)
    {
        Console.WriteLine("Addition of 2 nos: " + (n1 + n2));
    }

    MethodOverloading(int n1, int n2, int n3)
    {
        Console.WriteLine("Addition of 3 nos: " + (n1 + n2 + n3));
    }

    public static void Main(string[] args)
    {
        MethodOverloading obj = new MethodOverloading(43, 28);
        MethodOverloading obj1 = new MethodOverloading(43, 28, 37);
        Console.WriteLine("Himani T016");
    }
}
```

OUTPUT:

```
Hello! Delegate and Event Example.
Himani T016
```

F. Exceptional Handling:

CODE:

```
using System;
class Program
{
    static void Main()
    {
        try
        {
            int a = 10;
            int b = 0;
            int c = a / b;

            Console.WriteLine(c);
        }
        catch (DivideByZeroException)
        {
            Console.WriteLine("Cannot divide by zero.");
            Console.WriteLine("Himani T016");
        }
    }
}
```

OUTPUT: